

Closing corollaries and the completion of the grammar seabed (conditional §4 release)

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Abstract

Unit 289 closes Programme S with two thin corollaries and the seabed is marked complete as a conditional §4 release (Astra #36). (i) Different leading scales: if $(U_\ell/a_\ell, V_\ell/b_\ell) \Rightarrow (A, B)$ with deterministic nonzero normalisations and $P(B = 0) = 0$ then $(b_\ell/a_\ell) U_\ell/V_\ell \Rightarrow A/B$; assembled, with numerator target (μ_ϕ, j_ϕ) and denominator target (μ_1, j_1) , $N_\ell^{\mu_\phi - \mu_1} (\log N_\ell)^{j_1 - j_\phi} Z_\ell^0[\phi]/Z_\ell^0[1] \Rightarrow C_{\mu_\phi, j_\phi}^\phi(Y)/C_{\mu_1, j_1}^1(X)$. (ii) The conditional deterministic finite-chart normal-form expansion: for a fixed joint datum x and $Z_{\text{pop}} = \mathcal{Z}^{\text{glob}}(\cdot; x) + E$ with $E = o(N^{-\mu}(\log N)^j)$, the ordered normalised remainder of Z_{pop} tends to $C_{\mu, j}^{\text{glob}}(x)$, and with vanishing predecessors and a nonzero coefficient $Z_{\text{pop}}(N) \sim c N^{-\mu}(\log N)^j$ — a corollary of the assembly machinery, not the unrevised §3 theorem. The seabed now holds Headlines I–XXXVIII over units 1–289 with reviews v1–v31; no **sorry**, no additional **axiom**; 3416 jobs; pin f1c0138.

1 What “complete” means here

Astra #36’s wording is adopted verbatim in `HEADLINES.md`: the Lean formalisation establishes the fluctuation-function and ladder-algebra results, the Taylor-tree expansion in every positive normal dimension under the declared analytic hypotheses, a continuous-coefficient finite-cutoff stochastic extension through tangential integration and finite chart assembly, and the leading-order posterior quotient under joint data convergence, source predecessor cancellation, negligible decomposition residuals and an almost-surely nonzero limiting denominator coefficient. The stochastic results are conditional on convergence in the stated weighted- ℓ^1 /tangential-data topologies and on an externally supplied finite-chart decomposition; they do not derive those inputs from Hypothesis I, construct a resolution atlas, or identify a Gaussian empirical-process limit. All-orders posterior division and the population expansion in the unrevised §3 remain outside the release. “Complete” is a release decision about a delimited scope, not a claim that every paper label has a Lean counterpart.

2 The closing corollaries

The different-scale quotient is the generic quotient theorem applied to the normalised pair and the identity $(U/a)/(V/b) = (b/a)(U/V)$ for $a, b \neq 0$; the assembled form needs the pair of assembled remainders at two *different* targets, a two-target copy of the uniform-on-balls pair lemma. The deterministic bridge is the uniform-on-balls global remainder theorem evaluated at one point (`TendstoUniformlyOn.tendsto_at`), plus the residual and, for the equivalence, `isEquivalent_of_tendsto_one` (one explicit argument at this pin).

3 Hand-off

Reports: `normal-block-report-v3.pdf`, `taylor-tree-report.pdf`, `stochastic-taylor-tree-report.pdf` (all in `projects/grammar/staging/`); the annotated mirror `grammar_lean_annotated.tex` carries 111 statement-level links pinned at `f1c0138`. Last notes for the authors (Astra #36): adopt or amend the replacement statements for `prop:convergence` and `lemma:AsymInt`; separate §4.3 into conditional analysis and model verification; repair posterior statements at the source datum; state denominator nonvanishing independently; specify the algebra of all-orders division before requesting its formalisation; rewrite §3 before opening a geometry programme; freeze one authoritative release manifest.

4 Lessons of the programme

Nineteen units, five reviews, no mathematical defect found: the recipe was a frozen deterministic core, an abstract bookkeeping layer (finite-cutoff expansions), Mathlib’s convergence-in-distribution API used as-is, and a gate after every third unit. The recurring tooling lesson is mechanical: gate every commit on the build, the sorry audit and the line-length linter, and keep hypothesis names away from `variable` names.